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EU AI Act

Bulgaria

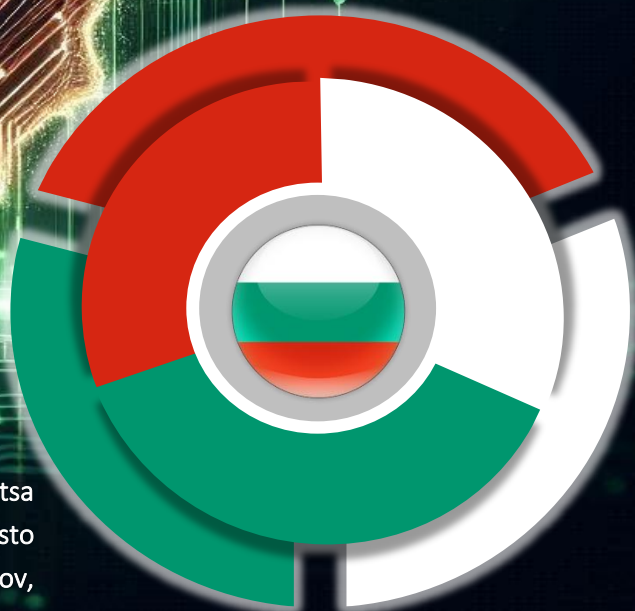
AI Ecosystem Brief

December 2025



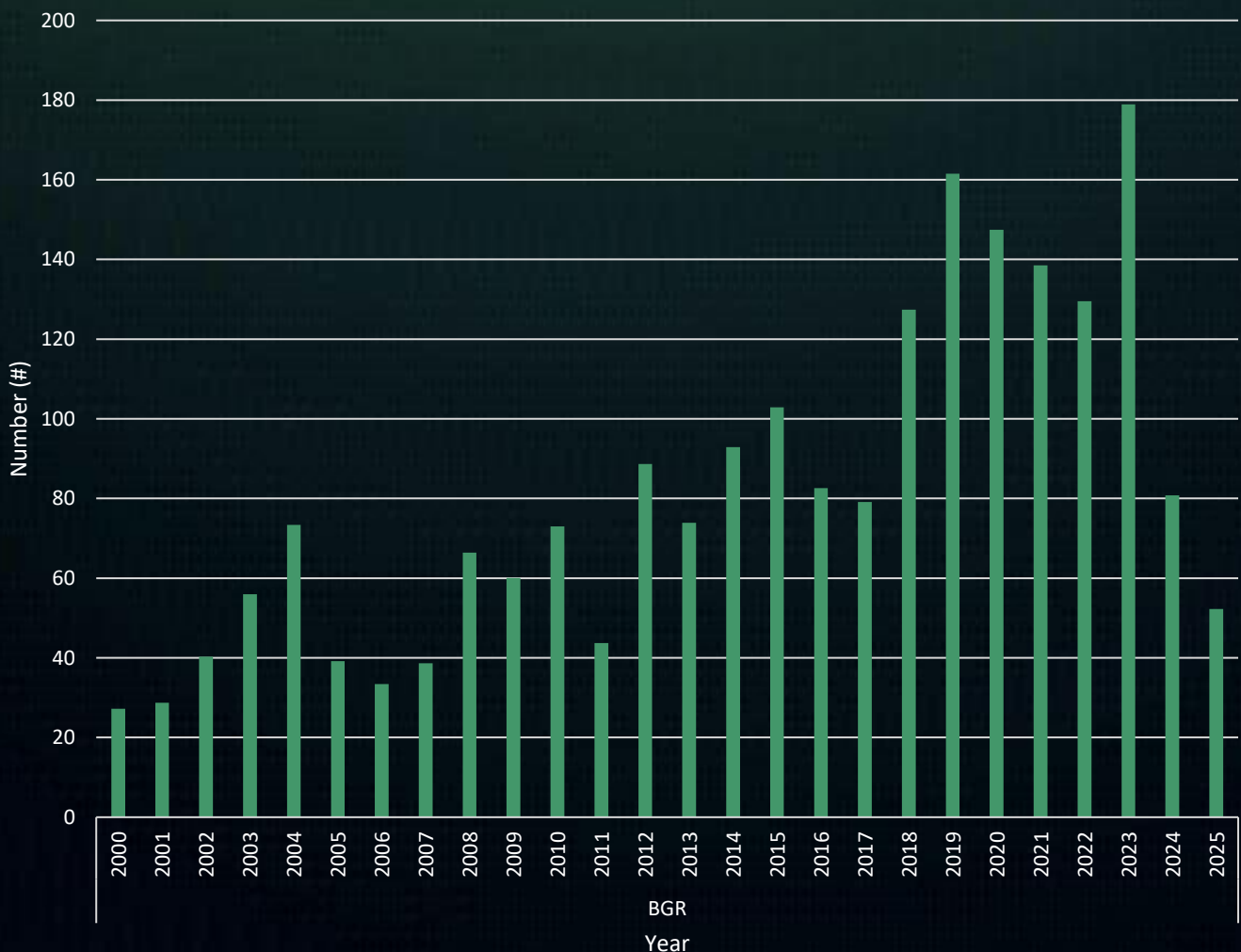
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Reinvigorating Bulgaria's AI Research Trajectory: Strategic Imperatives to Convert Volatile Output into Sustained Growth, Strengthen Innovation Capacity, and Secure Regional Competitiveness to 2030

*AI publications in Bulgaria over time
(November 2025)*



Explanation: The chart shows the number of AI publications by Bulgaria over time.

Bulgaria's AI research output (fractional contribution counts, 2000–2025) exhibits a long-term upward trend marked by significant volatility. Early levels in the 2000s remained modest (rising from 27.25 in 2000 to 73.38 in 2004), followed by alternating periods of growth and contraction through 2010. A more sustained acceleration begins after 2012, culminating in a sharp peak in 2019 (161.53), reflecting Bulgaria's increasing integration into European research networks, stronger participation in EU Framework Programmes, and the gradual effects of national digital modernisation strategies.

The period after 2019 reveals a plateau–decline–partial recovery cycle. Output falls from the 2019 peak through 2022 (129.52) before surging again to an all-time high of 178.95 in 2023 — likely tied to the establishment of new research institutions (including INSAIT), increased computational capability (supercomputers Discoverer and HEMUS), and early momentum around national language model development. However, the subsequent sharp decline in 2024 (80.76) and moderate rebound in 2025 (52.26 in the available data) indicate instability rather than sustained growth.

This volatility signals a research ecosystem undergoing transition — one that is expanding in capacity but still vulnerable to funding cycles, uneven institutional development, and limited private-sector absorption. Unlike larger EU Member States, Bulgaria's research surge is not yet fully anchored in a broad industrial innovation base. The data suggest that while Bulgaria can achieve high-output years driven by targeted initiatives, it lacks the structural stability needed for year-on-year continuity.

Several structural factors shape this pattern: dependence on competitive EU project funding; a still-maturing domestic R&D landscape; limited doctoral-level retention; and a comparatively small private sector engaging in AI research. Bulgaria's recent establishment of world-class research infrastructure positions it strongly for future growth, but scaling and stabilising output will require deeper linkages between academia, government, and industry.

Constructively, these trends offer Bulgaria a strategic moment. To stabilise AI research output and build long-term competitiveness, Bulgaria should:

1. **Strengthen applied research translation:** Create sustained funding lines linking INSAIT, GATE, universities, and industry clusters to convert research outputs — including language models — into commercial tools and sectoral solutions.

2. **Stabilise long-term R&D investment:** Reduce volatility by complementing EU-funded projects with national multi-year research programmes dedicated to AI, data science, and HPC-intensive experimentation.
3. **Deepen international research collaboration:** Expand co-publication and consortium participation across Horizon Europe and regional partnerships to increase visibility and smooth year-to-year variability in output.
4. **Invest in talent retention and expansion:** Broaden PhD scholarships, offer competitive contracts for early-career researchers, and leverage supercomputing access as a draw for international talent.
5. **Develop annual AI research monitoring:** Introduce a public AI research indicator series that tracks publication volume, collaboration networks, and impact metrics to guide strategic policy adjustments.

In summary, Bulgaria's AI research trajectory shows clear upward potential but is marked by significant year-to-year fluctuations. To transform episodic peaks into consistent, sustained growth, Bulgaria must couple its emerging scientific assets with stable funding, deeper industrial participation, strengthened talent pathways, and systematic monitoring. These steps will be essential for positioning the country as a resilient, high-impact contributor to European AI advancement by 2030.



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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

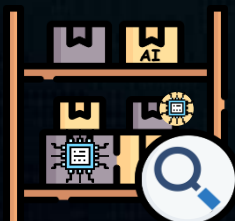
What Do We Do?

We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.

How Do We Do It?

Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.

Contents

This section outlines the structure of the report, detailing chapters on Bulgaria's AI policy landscape, regulatory initiatives, governance mechanisms, findings, and recommendations. It serves as a navigation guide, enabling readers to quickly locate specific analyses, datasets, and policy discussions relevant to Bulgaria's evolving artificial intelligence ecosystem.

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Overview

Artificial Intelligence (AI) is rapidly transforming societies and economies worldwide, and Bulgaria is positioning itself within this technological shift by aligning with the European vision of "reliable AI."

Embracing AI's potential while addressing emerging risks, Bulgaria has adopted a phased, sector-focused, and innovation-driven approach to AI governance. Rather than enacting standalone, comprehensive national legislation, Bulgaria is implementing the European Union's AI Act, while supplementing it with domestic frameworks, sectoral initiatives, and strategic tools to ensure responsible, secure, and transparent AI deployment.

At the core of Bulgaria's AI governance strategy is its alignment with the EU AI Act, which will apply from August 1, 2024, providing a harmonized regulatory baseline. At the national level, Bulgaria complements this framework with targeted initiatives, legislative proposals, and institutional support mechanisms. These tools are complemented by targeted sectoral efforts, including the implicit and explicit addressing of AI use in existing data protection laws, copyright laws, and e-administration frameworks, which influence AI development and use in Bulgaria. The legislative framework on national data assets also regulates the use of public data, including its potential use in AI.

Bulgaria's institutional AI governance structure is coordinated across multiple agencies. The government coordinates AI policy through its relevant ministries and agencies. The National Authority for Data Protection and Freedom of

Information (NAIH) plays a crucial role in enforcing data protection aspects relevant to AI. Other authorities overseeing related fields, such as those responsible for copyright and e-administration, also contribute to the oversight of AI systems within their respective purviews. An Interdepartmental Working Group is proposed to analyze the sector and prepare an operational National Plan/Roadmap.

Bulgaria's AI governance is also shaped by its commitment to international standards, particularly aligning with EU and OECD standards to facilitate consistency in AI governance and technology deployment. The country's AI strategy, which aligns with broader European values, sets out principles for ethical AI development and deployment.

As outlined in its Concept for the Development of Artificial Intelligence in Bulgaria Until 2030, Bulgaria envisions AI as a critical enabler of digital transformation, smart growth, and a prosperous democratic society. Strategic priorities include building reliable infrastructure for AI development, developing research capacity, creating knowledge and skills for AI development and use, supporting innovation for AI implementation, raising awareness and building trust in society, and establishing a regulatory framework for reliable AI in accordance with international ethical standards. Ethical considerations—such as transparency, fairness, and accountability—are integrated into AI design and deployment, even in the absence of a national, horizontal AI law.

Through a combination of EU harmonization, national infrastructure development, and alignment with global engagement, Bulgaria is advancing a governance model that balances technological leadership with democratic accountability. This report explores Bulgaria's evolving AI policy architecture, regulatory implementation strategies, and institutional innovations—highlighting how Bulgaria is shaping a future of trustworthy, inclusive, and resilient AI.



Specific AI

Governance or Law

Bulgaria lacks standalone AI legislation but follows a hybrid model combining the EU AI Act and sector-specific national laws. Existing data protection, copyright, and e-administration laws govern AI use. National initiatives support ethical AI aligned with EU values, ensuring oversight, transparency, and responsible deployment across public and private sectors.

Specific AI Governance in Bulgaria

Rather than enacting standalone, comprehensive national legislation, Bulgaria has embraced a hybrid governance model that combines EU-level regulation with domestic sector-specific initiatives. Central to this strategy is the alignment with the EU AI Act, which will apply from August 1, 2024, providing a harmonized regulatory baseline across the European Union. At the national level, Bulgaria complements this framework with targeted initiatives, legislative proposals, and institutional support mechanisms. Currently, Bulgaria does not have a specific national law dedicated solely to AI regulation. However, the use of AI is implicitly or explicitly addressed by various existing laws. These include data protection laws, which are technology-neutral but significantly impact AI operations due to AI's reliance on data. Copyright laws also apply, covering rights related to AI components (e.g., databases used for training) and AI-generated elements (e.g., software, creations). The legislative framework on national data assets regulates the use of public data, including its potential use in AI. Furthermore, the legislative framework on e-administration contains explicit rules on electronic administration tasks and activities conducted with automated means, which

often involve AI. These measures contribute to transparency obligations and introduce mechanisms for oversight of AI systems. Bulgaria's AI strategy, which aligns with broader European values, sets out principles for ethical AI development and deployment.

AI Ethics Guidelines and Sectoral Governance Tools

In the absence of AI-specific national legislation, Bulgaria leverages a constellation of sectoral frameworks and evolving guidance to uphold ethical and legal standards in AI. Key developments include: The National Authority for Data Protection and Freedom of Information (NAIH) plays a crucial role in enforcing data protection aspects relevant to AI. Other authorities overseeing related fields, such as those responsible for copyright and e-administration, also contribute to the oversight of AI systems within their respective purviews. Existing legislation, including data protection laws, copyright laws, and e-administration frameworks, implicitly or explicitly deal with the use of algorithms or AI, influencing their development and use in Bulgaria. These instruments collectively ensure that ethical considerations—such as transparency, fairness, and accountability—are integrated into AI design and deployment, even in the absence of a national, horizontal AI law.

Proposed AI Regulatory Authority and Future Outlook

Bulgaria has positioned itself as a strategic contributor to EU-wide AI policy while also investing in national oversight and leadership. The government coordinates AI policy through its relevant ministries and agencies. While specific names for coordinating committees or observatories are not explicitly detailed, the overarching AI strategy outlines a framework for tracking implementation and ensuring responsible AI deployment. The National Authority for Data Protection and Freedom of Information (NAIH) is expected to play a central role in regulatory oversight, enforcement, and guidance, especially in relation to privacy-preserving AI development. Looking forward, while Bulgaria does not intend to implement standalone AI legislation, its sector-specific legal amendments, coupled with institutional frameworks, public funding programmes, and alignment with the EU AI Act, indicate a clear national commitment to fostering trustworthy, ethical, and competitive AI innovation. National discussions will continue on the application of existing laws to new AI use cases, particularly concerning data ethics, intellectual property, and the broader societal impact of AI.

Bulgarian National AI Strategy

Bulgaria's AI governance strategy, outlined in its "Concept for the Development of Artificial Intelligence in Bulgaria Until 2030," is anchored in the European vision of "reliable AI." This approach emphasizes technological progress alongside a legal and ethical framework to ensure citizen security and rights. The strategy is operationalized through a proposed Interdepartmental Working Group, fostering collaboration across key state institutions, academia, business, and NGOs. It is embedded within broader digitalization initiatives, promoting responsible, secure, and inclusive AI by mobilizing stakeholders around shared goals.

Bulgaria's AI strategy reflects a comprehensive, ecosystem-oriented approach, prioritizing the development and implementation of AI across various sectors. Similar to broader European initiatives, Bulgaria emphasizes "trustworthy AI," ensuring ethical standards and minimizing harm. While the document doesn't specify an investment amount comparable to Flanders' €30 million, it outlines a commitment to significant financial provision through various national and European funding mechanisms.

The strategy aligns closely with European initiatives like the Digital Europe Programme, Horizon Europe, and the EU's focus on a "high-performance ecosystem" across the AI value chain.

The Bulgarian AI strategy, consistent with European directives, comprises several key areas of impact: (1) building a reliable infrastructure for AI development, including data infrastructure; (2) developing research capacity for scientific excellence; (3) creating knowledge and skills for AI development and use; (4) supporting innovation for AI implementation; (5) raising awareness and building trust in society; and (6) establishing a regulatory framework for reliable AI in accordance with international ethical standards. The document acknowledges that Bulgaria currently has a relatively lower growth in AI application implementation compared to some EU leaders but possesses qualified specialists to expand and deepen its use of AI technologies.

The emphasis on explainability, transparency, and data governance within the Bulgarian context reflects its ambition to ensure ethical deployment and contribute to global standards for trustworthy AI.

Science and
Scientific
Infrastructure

Education and
Skills

Intelligent
Industry

Electronic
Public
Administration

Sustainable
Agriculture

e-Health

Approach to AI Regulation

Bulgaria has adopted a compliance-focused, EU-aligned approach to AI governance, centred on harmonisation with the EU AI Act, strategic digital development, and the protection of fundamental rights. While national legislation directly regulating AI remains nascent, Bulgaria actively contributes to EU-level policymaking, institutional coordination, and infrastructure development to support safe and ethical AI deployment.

1. Regulatory and Strategic Frameworks

Bulgaria's AI governance is grounded in the European regulatory landscape, with no standalone national AI legislation in force. The EU AI Act, directly applicable across Member States, provides the primary legal framework. As of February 2025, Chapters I and II—covering scope, definitions, and prohibited AI practices—apply in Bulgaria, with full implementation set for August 2026. In response, Bulgaria is aligning its domestic governance with the EU AI Act. While a specific national AI law does not exist, the use of AI is implicitly or explicitly addressed by various existing laws. These include data protection laws, which are technology-neutral but significantly impact AI operations due to AI's reliance on data. Copyright laws also apply, covering rights related to AI components (e.g., databases used for training) and AI-generated elements (e.g., software, creations). The legislative framework on national data assets regulates the use of public data, including its potential use in AI. Furthermore, the legislative framework on e-administration contains explicit rules on electronic

administration tasks and activities conducted with automated means, which often involve AI. These measures contribute to transparency obligations and introduce mechanisms for oversight of AI systems. Bulgaria's AI strategy also aligns with broader European values for ethical AI development and deployment.

2. Institutional Support and Strategic Missions

Bulgaria's AI policy and implementation are coordinated through relevant ministries and agencies. The National Authority for Data Protection and Freedom of Information (NAIH) plays a crucial role in enforcing data protection aspects relevant to AI. Other authorities overseeing related fields, such as those responsible for copyright and e-administration, also contribute to the oversight of AI systems within their respective purviews.

3. Standards Development and Global Engagement

Bulgaria aligns its frameworks with EU and OECD standards, facilitating consistency in AI governance and technology deployment.

4. Sectoral and Extended Regulatory Approaches

AI governance in Bulgaria relies on the adaptation of existing legal frameworks, particularly in data protection, labor, IP, and civil liability. While no sector-specific AI laws exist, existing legislation implicitly or explicitly deals with the use of algorithms or AI. Product liability may be considered suitable for harm caused by AI technology, but it mainly covers death, personal injury, and property damages, potentially overlooking harm such as invasion of privacy, discrimination, or infringement of dignity. Challenges include establishing the source of harm due to interconnected and autonomous AI devices, and identifying whether defects stem from assignment, design, or development.

5. Fostering Innovation and Indigenous Development

The Bulgarian government is actively promoting digitalization and the use of AI. For instance, the Ministry of e-Government's project "Cloud e-services for the Administration" aims to automate work processes in central, regional, and municipal administrations. This project will integrate with centralized e-Government systems and specialized systems, creating a catalogue of automated e-Government services. The government is also developing an Information System for Centralised Construction and Maintenance of Registers, to automate and optimize processes, reduce data duplication, and enhance interoperability.

6. AI in Public Sector Transformation

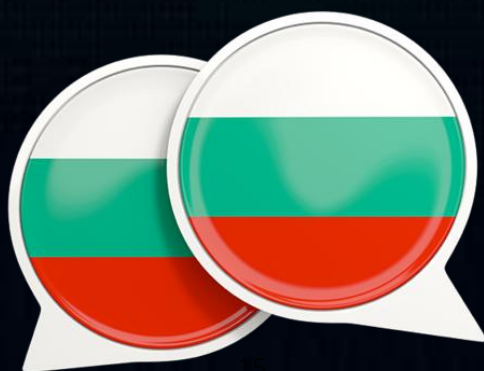
The Bulgarian government is actively exploring AI applications in public service delivery, with goals including administrative efficiency, data-driven policymaking, and service optimization. Automated services are delivered through an interactive government portal, enabling communication across state databases. Examples include services related to urban planning, business operating permits, funeral services, and public transport payments. However, the primary administrative laws, including the Administrative Judicial Review Act and the Administrative Code, currently lack specific provisions on the use of AI and algorithmic processes in public administration.

7. AI Skills, Education and Inclusion

While the provided content does not detail specific AI skills, education, and inclusion initiatives for Bulgaria, the broader context of Central and Eastern European countries suggests an ongoing need for competence development. Other regional examples show efforts to revise school curricula to introduce AI and coding, and the implementation of digital systems in service sectors (e.g., Teacher Portal, automated State exams).

8. Infrastructure, Compute and Data Foundations

To support AI growth, Bulgaria is enhancing its data and digital infrastructure. The Minister of e-Government oversees the Register of Registers, which catalogues the registers and databases of primary data controllers. An Electronic Messaging Environment (EME) has been implemented to streamline document management within public administration, and a Secure Electronic Service System is provided for administrations without technical resources. The government is also developing an Information System for Centralised Construction and Maintenance of Registers to enhance interoperability, control, and monitor data access and dissemination.



Approach to Foster Innovation

Bulgaria's approach to artificial intelligence is rooted in aligning with the European vision of "reliable AI," emphasizing technological progress alongside a robust legal and ethical framework to ensure citizen security and rights. The "Concept for the Development of Artificial Intelligence in Bulgaria Until 2030" outlines a strategic plan for AI development and implementation, focusing on smart growth and a prosperous democratic society. This strategy leverages a collaborative model, proposing an Interdepartmental Working Group to coordinate efforts across key state institutions, academia, business, and non-governmental organizations, to embed AI within broader digitalization initiatives.

Unlocking Economic Value through Targeted AI Deployment

Bulgaria's strategy identifies "AI consuming sectors" and "AI developing sectors" as key areas for AI adoption. Priority sectors are listed with arguments for their importance. The strategy encourages AI deployment through innovation support, including joint research and applied projects, and the establishment of start-ups

and technology transfer to industry, particularly SMEs. Examples of potential AI applications in public services include chatbots, predictive analytics, and computer vision. The document also highlights AI's potential in healthcare (e.g., disease prevention, medical image analysis) and intelligent agriculture (e.g., efficient land use and automation).

Catalysing Innovation through Strategic Clustering and Ecosystems

The Bulgarian AI ecosystem aims to overcome fragmentation among research units by creating conditions for building human potential within a connected national academic environment. This includes creating a Bulgarian Research Programme for comprehensible, people-oriented AI and robotics, uniting scientific organizations and universities with original developments. The strategy emphasizes strengthening and developing capacity for research and technology transfer in AI and related technologies, focusing on areas with strong Bulgarian expertise like neural networks, machine learning, hybrid AI, and natural language processing.

Enabling Trusted and Responsible Data Use

Bulgaria fosters secure and ethical AI through adherence to GDPR principles and by emphasizing data governance. The strategy explicitly references the EC's seven key requirements for reliable AI, which include privacy and data governance, ensuring data is free from inaccuracies and bias. Measures include encouraging the sharing of non-personal data through ethical and secure exchange scenarios, creating rules for anonymization and pseudonymization of personal data (e.g., in healthcare), and incentivizing organizations to share data in the national Open Data Portal. The strategy also focuses on providing accessible, high-quality data for machine self-learning and building knowledge graphs.

Scaling AI in the Public Sector

The Bulgarian public administration is seen as a crucial testbed for AI applications. The strategy suggests AI implementation in public services through initiatives that can include predictive services, data analytics, and intelligent automation for citizen services. Examples include chatbots for public information, AI-powered systems for analyzing traffic, and optimization of energy systems and utilities in smart cities. The government aims to accelerate the adoption of trustworthy AI to improve service efficiency and address societal challenges.

Advancing Compute and Research Infrastructure

Bulgaria plans to invest in modern communications and scientific infrastructure for AI development, including high-performance and distributed computing. This includes leveraging sites like the National Center for High-Performance and Distributed Computing (NCHDC) and the European Supercomputer Complex in Sofia Tech Park. The strategy aims to provide high-tech AI platforms integrated with European ones to ensure interoperability and open access for developing innovative AI solutions and services. It also mentions constructing digital information hubs in AI and robotics, co-financed by the "Digital Europe" program.

Empowering SMEs and Startups

Support for innovation to implement AI in practice, particularly for SMEs, is a core priority. The strategy outlines actions to stimulate joint research and applied projects, establish start-ups from academic organizations, and facilitate technology transfer to industry. The concept emphasizes that Bulgaria should focus on technological specialization in the field of data economy, as it may be challenging to achieve strong industrial specialization without a critical mass of leading AI companies.

Developing a Future-Ready Workforce

Bulgaria addresses the digital skills gap through an integrated education and lifelong learning strategy. The focus is on strengthening capacity for research and technology transfer. The strategy highlights the importance of "hardskills" (mathematics, data analysis, programming), STEM skills, digital skills, and transversal skills, emphasizing their development at all levels of education.

Aligning AI Innovation with Sustainability Goals

While not explicitly detailing "green transition" initiatives with AI, the strategy touches upon areas that align with sustainability. For example, it mentions AI's role in ecology and the environment, including image recognition for biodiversity observation, pollution analysis, and predicting environmental factors. AI systems for integrated processing of unstructured data are envisioned to inform about environmental crises and offer optimal solutions for environmental sustainability.

Digital Decade

2025: Report

Bulgaria is advancing initiatives expected to increase the use of AI across its business sector. A notable example featured in the 2024 Digital Decade report is BgGPT, an open-source AI model tailored to the Bulgarian language, developed by the INSAIT research institute and launched in 2024.

In March 2024, the country was selected to host BRAIN++, an AI training centre located at Sofia Tech Park. This project forms part of a broader EUR 485 million European Commission initiative to establish seven AI factories across the EU. Construction of the high-performance GPU AI data centre is scheduled to begin in 2026. With an estimated cost of EUR 90 million—half of which is covered by the Bulgarian government—BRAIN++ is a joint initiative between Sofia Tech Park and INSAIT. The facility will become part of the EU's supercomputing infrastructure, facilitating the exchange of computational resources, datasets, and expertise throughout Europe and contributing to a responsible, ethical AI ecosystem.

It will also offer the BulgAI Sandbox, a secure cloud-based environment designed for SMEs and researchers to develop, test, and deploy AI solutions. This hybrid model integrates Bulgaria's Discoverer+ EuroHPC supercomputer with flexible cloud capabilities, enhancing accessibility for innovation in AI.

The Bulgarian government is actively tracking AI maturity levels within the enterprise sector, classifying companies into four categories. The 'leaders' group, representing 14% of firms, are highly innovative and effectively integrate AI into

their operations with dedicated budgets. The 'catch-up' group, covering 23%, are fast adopters experimenting with various AI tools—often in warehouse automation and AI-based management—though they lag slightly in innovation. The largest segment, the 'laggards' at 41%, demonstrates limited AI use and a weak grasp of its benefits, engaging with AI only occasionally and on a superficial level.

Meanwhile, 22% fall into the 'I Want To' category—businesses enthusiastic about AI but hindered by inadequate implementation processes and internal capabilities despite having set aside funding.

The 2024 recommendation for Bulgaria was to expand on its participation in the ALT-EDIC and capitalise on assets like BgGPT to introduce new initiatives aimed at cultivating the AI ecosystem and supporting widespread adoption.

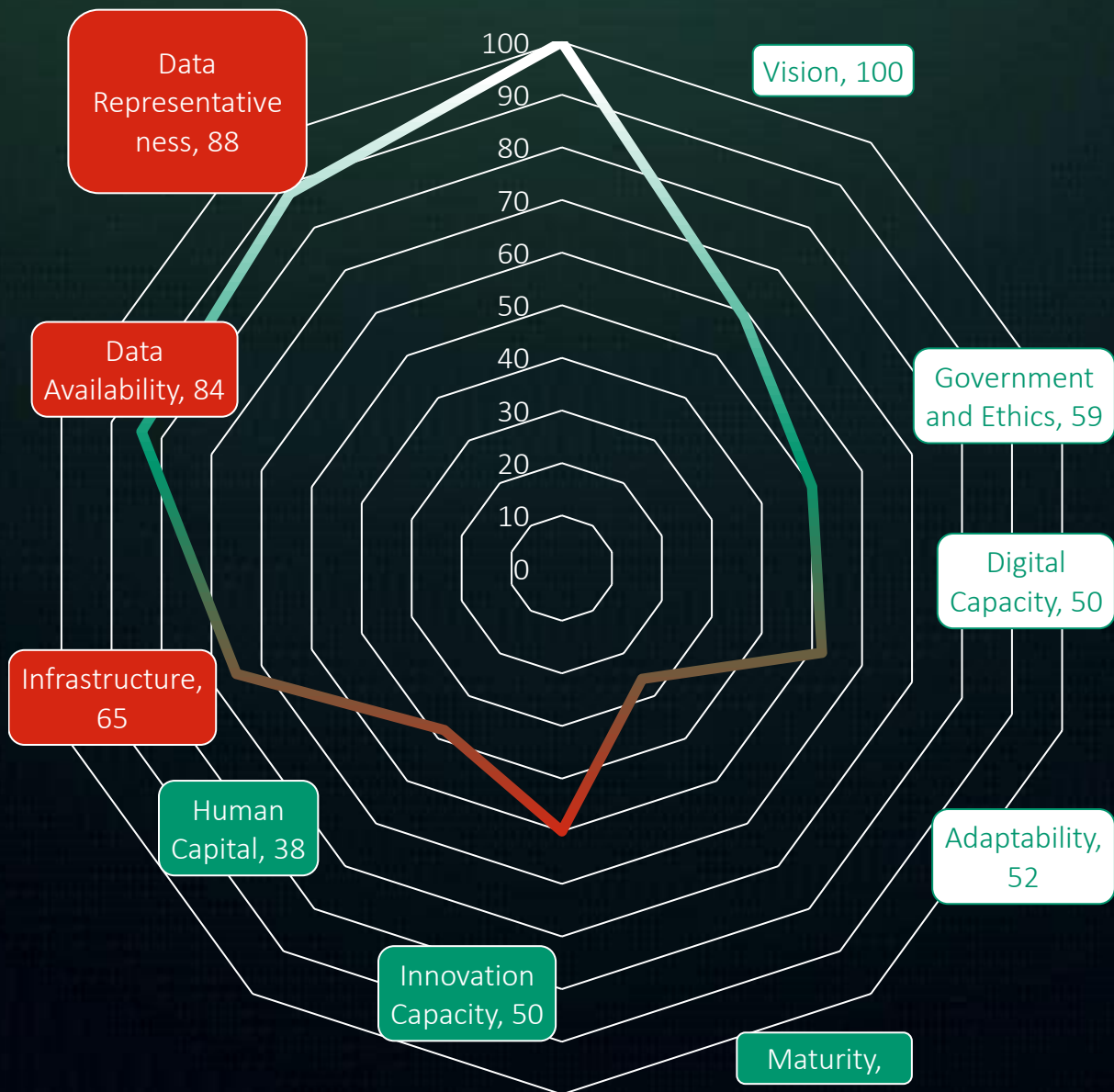
Although no new initiatives were launched in 2024, Bulgaria continued to carry out previously planned measures. Collaborative structures such as the Advisory Council of the ICT Community and the thematic working group on "AI – Guarantees of equal access and respect for human rights," established by the Ministry of Electronic Governance (MEG), are contributing to stronger links among government, civil society, and the ICT sector. These platforms promote the exchange of best practices, foster innovation, and facilitate discussions on ethical and policy-related aspects of AI.

Bulgaria's roadmap under the Digital Decade includes targeted AI measures, with integration across sector-specific strategies enabling a more effective and balanced allocation of resources. Additionally, a proposal has been submitted under the European Commission's Technical Support Instrument to enhance administrative capacity for AI governance, particularly in line with the forthcoming EU AI Act and the Council of Europe's Convention on AI, Human Rights, Democracy, and the Rule of Law.

Government AI Readiness

‘Eastern Europe ranks 4th in this year’s Government AI Readiness Index, with an average score of 57.88—over 10 points above the global average (47.59). Estonia (72.62) leads the region and is the only Eastern European country to feature in the global top 25. The region performs above the global average across all three pillars. Its strongest performance is in the Data & Infrastructure pillar (72.10), making it the third-best performing region globally in this area. Eastern Europe also shows solid results in the Government pillar (62.52), where it again ranks third worldwide. However, the Technology Sector remains a challenge, with a regional score of 39.03—more than 20 points behind the other two pillars—and ranking fifth globally, falling narrowly behind the Middle East and North Africa (39.34). This reflects the need for further investment in technological capacity and innovation to unlock the region’s full AI readiness potential.

2024 Scores by Dimension



Government

Bulgaria scores 65.19 in the Government pillar, reflecting a strong performance where it ranks third worldwide. This indicates robust governance frameworks across the region, with a focus on preparing for AI implementation and aligning with broader European principles. While the input doesn't detail specific regional initiatives beyond "strong governance frameworks," the consistent performance suggests a concerted effort by governments in the region to establish policies and prepare for the integration of AI, likely influenced by the ongoing preparations for the EU AI Act and similar ethical and regulatory guidelines.

Technology Sector

Bulgaria scores 37.88 in the Technology pillar, ranking fifth globally, narrowly behind the Middle East and North Africa. This highlights the Technology Sector as a significant challenge for the region, with a score more than 20 points behind its performance in the other two pillars. This suggests a need for further investment in technological capacity, innovation, and R&D scaling. While the region boasts qualified specialists, accelerating venture capital availability, supporting startups, and fostering commercialization efforts will be critical for Eastern Europe to unlock its full AI readiness potential and bridge this gap.

Data & infrastructure Sector

Bulgaria achieves an impressive 78.85 in the Data & Infrastructure pillar, making it the third-best performing region globally in this area. This strong performance is driven by robust data infrastructure. The region benefits from developments in digital connectivity and an increasing focus on secure data environments. This foundational strength positions Eastern Europe as a leader among emerging regions in its ability to support and leverage data for AI development and deployment.

Bulgaria represents a leading country among emerging economies in AI readiness, balancing robust governance and strong data infrastructure with a developing technological sector. Its strong performance in the Government and Data & Infrastructure pillars drives its overall progress, while continued investment and focus on the Technology Sector will be essential for the region to maximize its AI potential

Conclusion

Bulgaria's Artificial Intelligence (AI) ecosystem is at a pivotal moment, working to translate its foundational strengths—particularly in government readiness and data infrastructure—into tangible and sustainable impact. Guided by the "Concept for the Development of Artificial Intelligence in Bulgaria Until 2030" and steadfastly aligning with the European vision of "reliable AI" and the forthcoming EU AI Act, Bulgaria is building a framework for AI innovation that embraces democratic accountability, technological sovereignty, and human-centric principles. Its strategic priorities, which include robust infrastructure development, research capacity building, and a strong emphasis on ethical guidelines, demonstrate Bulgaria's commitment to leveraging AI for smart growth and a prosperous democratic society.

However, the journey to a fully mature AI ecosystem presents distinct challenges. As indicated by its lower score in the Technology Sector, Bulgaria must accelerate the transfer of AI research into scalable, commercial applications, particularly within its SME landscape. Addressing talent shortages, enhancing access to advanced compute resources, and fostering a more dynamic venture capital environment are critical next steps to unlock the region's full AI potential. While Bulgaria's governance model benefits significantly from its proactive alignment with EU harmonization and strong regulatory foresight, domestic implementation must continue to prioritize clarity, administrative simplicity, and effective enforcement. The proposed Interdepartmental Working Group and the ongoing role of institutions like the National Authority for Data Protection and Freedom of Information (NAIH) will be central to shaping an operational landscape that builds public trust in AI deployments.

To secure long-term leadership and maximize the societal benefits of AI, Bulgaria must continue to invest in inclusive innovation strategies that empower local enterprises, bridge regional divides, and foster deeper collaboration across academia, startups, and established industries. Its strategic international engagement, primarily through adherence to EU and OECD standards, positions Bulgaria to contribute meaningfully to global AI norms and governance frameworks, particularly around safety, ethics, and sustainability. If Bulgaria maintains its strategic coherence, technical ambition, and strong societal mandate, it is well-positioned to not only lead among emerging economies in AI readiness but also to define a trajectory for responsible, human-aligned AI systems within the broader European context.

**The strategic groundwork is
clearly laid—the moment to
accelerate comprehensive
execution is now.**

Appendix A: OECD AI Progress Report

This note summarises the OECD country note on Bulgaria’s progress implementing its 2019–20 AI strategy (the “Concept for the Development of Artificial Intelligence in Bulgaria until 2030”, often referred to as Bulgaria’s national AI strategy) and how those commitments are being integrated into broader national data, research and digital skills initiatives.

Bulgaria’s national AI strategy (adopted in 2020) sets out a broad, six-pillar vision that spans infrastructure, education, research, data potential, sectoral innovation and ethics. This strategic framing has led to concrete investments in computing capacity, new research institutions and targeted sectoral pilots, but the strategy has so far lacked a time-bound action plan with clear implementation milestones — leaving coordination, sequencing and measurable delivery as near-term challenges. Key recent advances include strengthening HPC capacity (two operational supercomputers and a third under development), the creation of INSAIT (which has accelerated research and local model development such as BgGPT), and a growing ecosystem of EDIHs, clusters and funding instruments aimed at SME uptake. Together these developments show a country that is building technical and institutional foundations quickly, but which must now translate strategy into routinised delivery, scale pilots into broad market adoption, and operationalise data governance and accountability mechanisms.

Strengths

- Strategic, multi-pillar vision: The 2020 national AI concept provides a coherent list of priorities (infrastructure, education, research, data, sectoral innovation, ethical AI) giving policymakers a single reference for national AI efforts.
- Growing HPC and data capacity: Bulgaria has significantly upgraded computing infrastructure — Discoverer and HEMUS are operational and Avitohol is under development — and is putting in place a National Data Strategy (2025–2030) and a National Data Governance Reference Framework (NDGRF). The Big Data for Smart Society Institute (GATE) is positioned as both a centre of excellence and the country's International Data Spaces Association Hub.
- Research lift and local language models: The Institute for Computer Science, Artificial Intelligence and Technology (INSAIT), established in 2023, strengthens national research capacity and supported the creation of BgGPT — an open-source Bulgarian language model — signalling capability to develop locally-relevant models and tools.
- Skills and education pipeline: Bulgaria has launched AI guidelines for schools, a specialised AI programming course piloted in secondary schools, and INSAIT Explorer to link university study with research experience; 2024 operations to boost digital and new skills aim to widen workforce readiness.
- Support for market adoption and SMEs: An active AI Cluster Bulgaria, a network of European Digital Innovation Hubs (EDIHs) and the Competitiveness and Innovation in Enterprises programme (which includes ~EUR 200 million in support instruments) are in place to help firms — especially SMEs — adopt AI. Sectoral initiatives (VELES in health, INNOAIR for urban mobility, AgroHub.BG for agriculture) give practical entry points for uptake.

Constraints and risks

- Strategy vs. delivery gap: The national concept is comprehensive but lacks a detailed, time-bound action plan with assigned responsibilities and milestones — increasing the risk of fragmented implementation and missed targets.

- Data governance still maturing: While a National Data Strategy and NDGRF are being developed, operational rules, standard APIs or certified intermediaries for safe cross-institutional data sharing are not yet fully established — limiting large-scale, trusted data reuse for AI.
- Scaling research into the market: INSAIT and BgGPT show research strength, but turning lab successes into sustainable commercial products and services (particularly from SMEs) will require stronger commercialization pathways and financing match-making.
- Talent bottlenecks and retention: Early education and upskilling steps are positive, yet demand for specialised AI, data-engineering and HPC skills risks outpacing supply unless reskilling, international attraction and retention measures accelerate.
- Transparency and measurable outcomes: There is limited public reporting on deployment outcomes, impact assessments or enforcement actions tied to AI systems — constraining public accountability and trust as deployment expands.

Practical recommendations (constructive next steps)

1. Convert the strategy into a time-bound action plan: Publish a clear implementation roadmap (owner, budget, milestones) for the 2025–2030 period so ministries, research centres and industry have aligned delivery targets and accountability mechanisms.
2. Operationalise data governance and trusted access: Fast-track the NDGRF and National Data Strategy into practical delivery (standard APIs, certified data intermediaries, legal templates for data pooling and re-use) and leverage GATE as a practical hub for cross-institutional data projects.
3. Scale SME pathways and market pull: Link EDIHs, the Competitiveness & Innovation programme and Invest-style financing with public procurement pilots and matched commercialization grants to help pilots (including BgGPT-based tools) move into wider market use.
4. Strengthen talent pipelines and HPC access: Couple expanded training and retraining programmes with clearer pathways to HPC time (via Discoverer/HEMUS/Avitohol) for researchers and firms; consider targeted fellowships and international researcher attraction to close short-term skills gaps.

5. Publish regular impact and sectoral guidance: Issue public progress reports on AI deployments, and fast-track pragmatic sectoral guidance (healthcare, agriculture, mobility) and certification routes so organisations can adopt validated tools safely and transparently.

Overall assessment: The OECD country note paints Bulgaria as a country that has rapidly built core technical and institutional building blocks (HPC capacity, INSAIT, national data frameworks in development, local language model work and an active innovation ecosystem) but that now needs to pivot from strategy and pioneering pilots to structured implementation: time-bound action plans, operational data governance, SME commercialization pathways, transparent reporting and accelerated skills/HPC access will be the levers that turn Bulgaria's clear strategic vision into broad, trustworthy and sustainable AI adoption.



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This section compiles all primary and secondary sources cited—legal texts, policy papers, academic studies, and official reports. It provides a transparent basis for the report’s analyses and recommendations, enabling verification, further research, and continued dialogue on AI policy and governance in Bulgaria.

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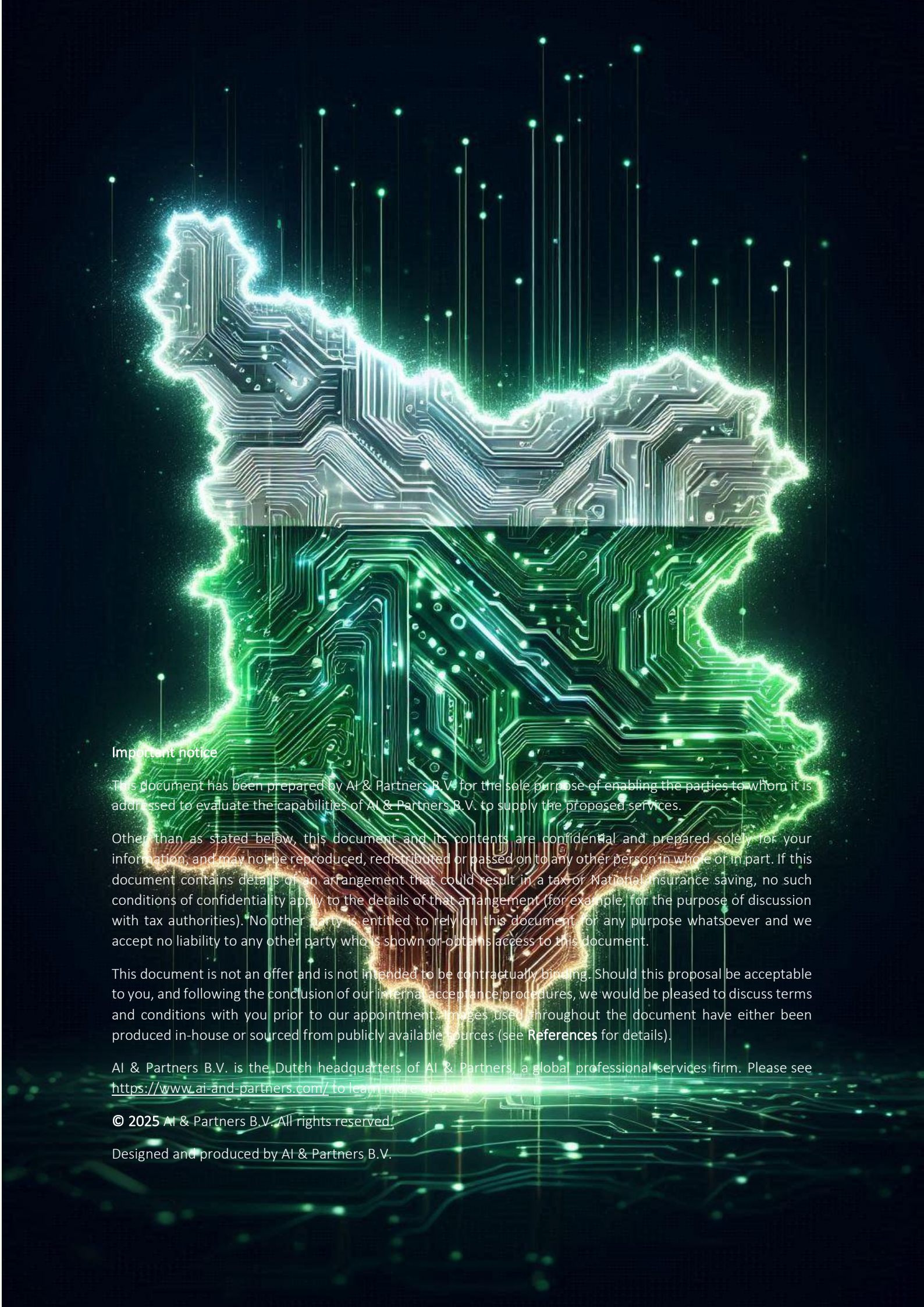
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